

August 21, 2026

Dear BigGeo Hiring Team,

I am writing to apply for the Spatial Data Onboarding Specialist position at BigGeo. My M.Sc. research at the University of Calgary was already closely connected to BigGeo through a BigGeo/Mitacs research software role, where I worked on scalable management of Earth observation data. Because of that experience, this role feels like a very natural next step: preparing, validating, documenting, and integrating spatial datasets so they can become reliable infrastructure for real-world spatial intelligence.

In my research software work, I built C++ and Python systems for processing, encoding, storing, and retrieving satellite-imagery data. I automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs, then developed DGGS-based workflows that converted imagery into structured tensor files for multiresolution access. That work required careful attention to data quality, spatial structure, processing performance, metadata context, and whether downstream users could retrieve the right data reliably. I also built DEM data workflows using Rasterio, Google Earth Engine, Streamlit, and PyTorch, mainly focused on collecting, preprocessing, validating, and visualizing terrain datasets so model outputs and source data could be inspected more reliably.

I was also excited to see the posting call out AI-enabled workflow support. I have been building agentic AI workflows for job matching and application preparation, using LLMs to assist with structured review, decision support, documentation, and repeatable human-in-the-loop checks. I have also used AI tools to speed up scripting, debugging, review, and technical writing while still validating the outputs myself. That maps well to BigGeo's emphasis on using AI to accelerate schema discovery, metadata drafting, transformation scripting, anomaly detection, and onboarding tasks without treating AI as a substitute for accuracy.

Beyond geospatial research, I bring backend and data workflow experience with Django REST APIs, PostgreSQL, Git, Linux, Docker, AWS-based coursework, and GitHub Actions. These experiences give me a practical foundation for working with engineers, reasoning about APIs and platform behavior, documenting technical decisions, and supporting data workflows that need to be reliable rather than just functional.

What excites me most is the practical importance of the role. Spatial data is only useful when it is clean, structured, documented, indexed, and trusted. I would be grateful for the opportunity to bring my BigGeo-adjacent research background, spatial data engineering experience, and careful documentation habits into a role that directly shapes the quality and reliability of the Spatial Cloud data ecosystem.

Thank you for considering my application. I look forward to the opportunity to discuss how my geospatial software, remote-sensing, validation, and AI-assisted workflow experience can contribute to BigGeo.

Sincerely,
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